

Regression

It is a classification algorithm used when data is continuous

Types of regression:-

- 1) simple linear regression
- 2) Multiple linear ————
- 3) Polynomial regression.

1) Simple linear regression

Linear is a straight line

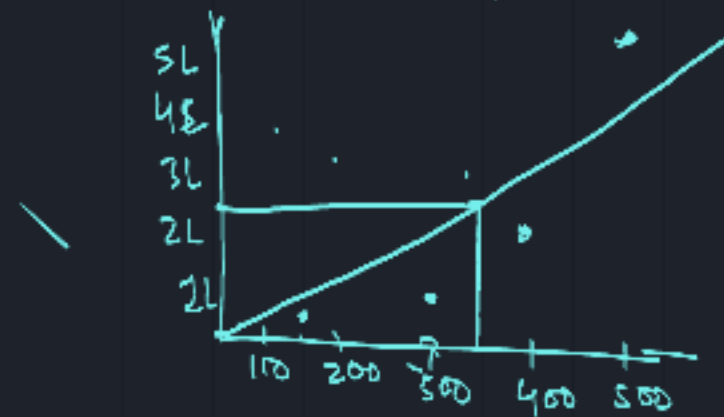
House Price

300 1L

400 2L

500 5L

150 50K



350 sq ft = 2,50,000

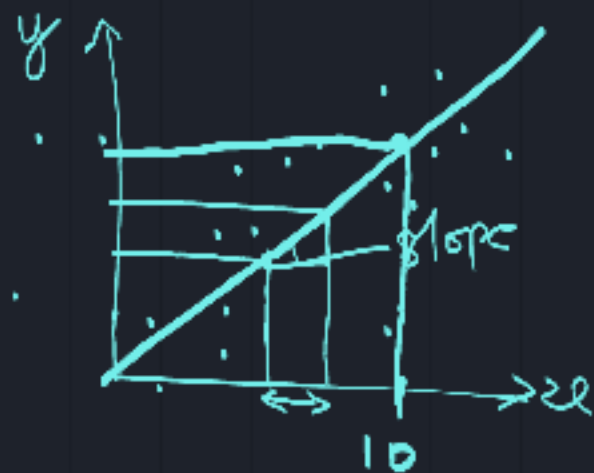
for simple linear regression we have to find best fit line

best fit line should be in such a way that all data point should near to straight line.

formula of straight line

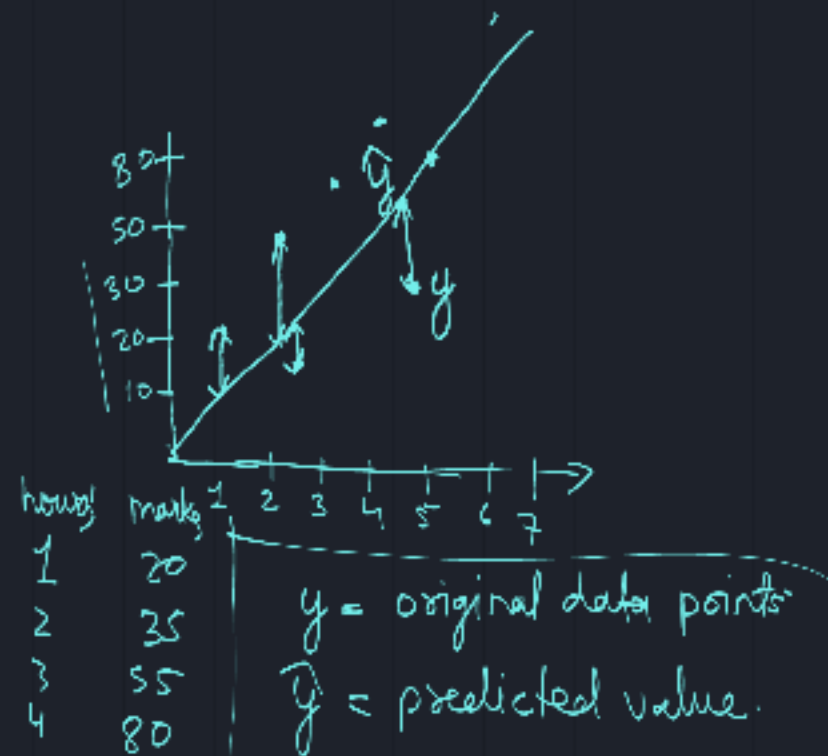
$$y = mx + c$$

slope intercept



$m(\text{slope})$ = when x changes by 1 unit
then the rate of change of y .
this distance is called as slope.

intercept = when x is 0 the intersection
of straight line on y -axis
that is called as intercept.



$$\text{Residual/error} = y_i - \hat{y}_i$$

$$\text{Residual sum of squares} = \sum (y_i - \hat{y}_i)^2$$

$$\text{mean squared error} = \frac{\sum (y_i - \hat{y}_i)^2}{n}$$

(Cost function)

straight line

$$y = mx + c$$

$$\hat{y} = \theta_1 x + \theta_0$$

$$J = \frac{\sum (y_i - (\theta_1 x_i + \theta_0))^2}{n}$$

$$\frac{d}{d\theta_1} \frac{\sum (y_i - (\theta_1 x_i + \theta_0))^2}{n} = 0$$

$$\frac{d}{d\theta_0} \frac{\sum (y_i - (\theta_1 x_i + \theta_0))^2}{n} = 0$$

(Ordinary Least square)

$$\theta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$